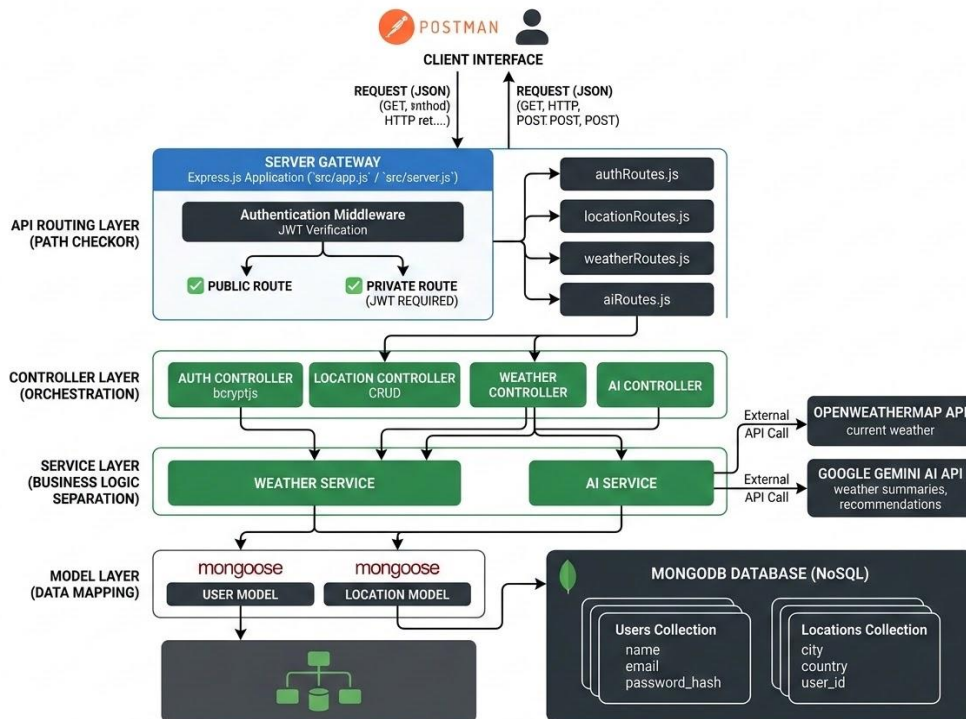


Phase 3

Project Design Phase

Phase 1: Project Design Phase



The AI WeatherWise application follows the MVC pattern to decouple business logic from data management. The Controller layer orchestrates weather fetching from OpenWeatherMap and generation of AI insights via Gemini, while the Model layer manages User and Location entities.

Components Definition

- **Interface:** API tools like Postman query distinct route paths to extract JSON entities.
- **Express Server Gateway:** Monitors network ports, maps explicit HTTP methods (GET, POST, PUT, DELETE), parses JSON payloads, and handles CORS headers.
- **Authentication & Role Matrix Middleware:** Extracts tokens from headers, validates user context signatures, and evaluates if the specific role satisfies authorization thresholds for guarded resource paths.
- **Granular Services (Logic Modules):** Enforces isolated domain rules for blog lifecycles (Draft, Pending Approval, Scheduled, Published), comment queues, and system reporting indicators.
- **Database Interface Layer:** Mongoose translates operational application logic into performant native MongoDB queries.
- **AI Service Integration:** Leverages `@google/generative-ai` to generate weather summaries and personalized recommendations based on live temperature and conditions.